Declaration of AI assistance

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Author-prepared disclosure; not an official publisher form or signature.

Manuscript. Even-intersecting families of lattice paths and the Catalan numbers

Human authors. Deep Bhattacharjee.

Claude through Claude Code assisted the original proof arguments, finite verification programs and drafting. OpenAI Codex assisted the induction and exceptional-pair audit, the nonnegative-coordinate range in the cell lemma, independent geometric-edge checks, references and typesetting.

Human oversight. The submitting author stated on 25 October 2026 that the text is a mix of human-written and AI-generated material and that all parts have been checked and understood by a human. This records that author declaration. Human authors retain responsibility for mathematical accuracy, citations and the final version; AI systems are not human coauthors.

Verification scope. The Lean checks cover stated finite ranges, including path-family classification for n at most 3. The all-length classification is justified by the written induction and is not claimed to be fully formalised in Lean.

Supporting record. The separate AI_PROVENANCE.md identifies the documented tasks and tools. No unprovided model version, human signature, confidential transcript or journal approval is asserted.

Policy reference. https://www.elsevier.com/about/policies-and-standards/generative-ai-policies-for-journals

Prepared on 25 October 2026. The author reviews this statement and completes any additional live-portal declarations before submission.